La Scuola d'Italia / Dr. Huson / IB Math 12 (a)
28 October 2026

Nickname:

2.6 Early Finishers: Trigonometry and Statistics

For students who finish the quiz early. A calculator is allowed. Give answers to three significant figures, and give angles in radians when the question gives an angle in radians.

1. [Maximum mark: 8]

   In triangle PQR, $PQ = 12\,\text{cm}$, $QR = 9\,\text{cm}$ and $Q\hat{P}R = 0.7$ radians.

   (a) Find the two possible values of $P\hat{R}Q$. [3]

   In the rest of this question, $P\hat{R}Q$ is obtuse.

   (b) Find $P\hat{Q}R$. [1]

   (c) Find PR. [2]

   (d) Find the area of triangle PQR. [2]

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2. [Maximum mark: 9]

The table shows the times, t minutes, taken by 40 runners to complete a race.

Time, t	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$	$50 \leq t < 60$
Number of runners	6	p	12	q	3

An estimate of the mean time is 31.25 minutes.

(a) Find the value of p and the value of q. [4]

(b) Find an estimate of the standard deviation of the times. [2]

Every runner is given a time penalty of 2 minutes, and the new times are then written in seconds.

(c) Find an estimate of the mean and of the standard deviation of the new times, in seconds. [3]

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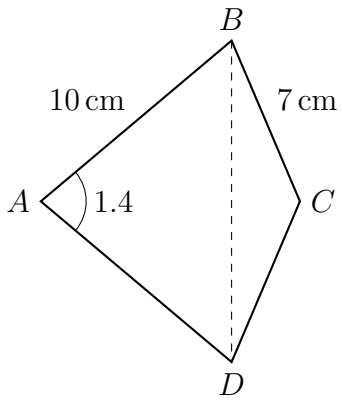
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2.6 Early Finishers: Trigonometry and Statistics

For students who finish the quiz early. A calculator is allowed. Give answers to three significant figures, and give angles in radians when the question gives an angle in radians.

1. [Maximum mark: 11]

   $ABCD$ is a kite with $AB = AD = 10\,\text{cm}$, $CB = CD = 7\,\text{cm}$ and $B\hat{A}D = 1.4$ radians.



   (a) Find BD. [3]

   (b) Find $B\hat{C}D$. [3]

   (c) Find the area of the kite. [3]

   (d) Find $A\hat{B}C$. [2]

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2. [Maximum mark: 10]

The table shows the number of hours of sunshine, x, and the number of cold drinks sold, y, at a beach kiosk on six days.

Hours of sunshine, x	2	4	5	7	8	10
Drinks sold, y	12	19	14	k	27	26

The mean number of drinks sold is 20.

(a) Find the value of k. [2]

(b) Find Pearson's product-moment correlation coefficient, r, and write down the regression line of y on x, in the form $y = ax + b$. [3]

(c) On another day, 24 drinks were sold. Use the appropriate regression line to estimate the number of hours of sunshine on that day. [3]

(d) Write down the coordinates of the point where the regression line of y on x and the regression line of x on y intersect. [2]